CLASS 9 SCIENCE · QUICK NOTES

CHAPTER 10

Sound Waves

Characteristics and Applications · Production · Propagation · Wave properties · Pitch, loudness, timbre · Echo · Ultrasound

THE CHAPTER IN 30 SECONDS

Sound is made by **vibrating objects** and travels as a **longitudinal mechanical wave** of compressions and rarefactions. It needs a **medium** (no sound in vacuum). Its speed is $\mathbf{v = \lambda \times \nu}$, and it is fastest in solids. We hear **20 Hz–20 kHz**. Pitch comes from frequency, loudness from amplitude, and quality (timbre) from the mix of frequencies. **Echoes** measure distances; **ultrasound** is used for sonar, imaging and cleaning.

1 Production and propagation of sound

NCERT 10.1, 10.2

- **Sound is produced by vibration**: a plucked string, a struck drum, a tuning fork (its prongs make air vibrate; a vibrating fork touching water makes splashes), our vocal cords.
- Sound moves through a **medium**: air, water, steel. Vibrating particles pass the disturbance to neighbours; the **particles do not travel** along, only the disturbance does.
- **No medium, no sound.** A bell ringing inside a bell jar becomes fainter as air is pumped out and is not heard in a vacuum (even though you can still see it vibrate).

2 Sound as a wave

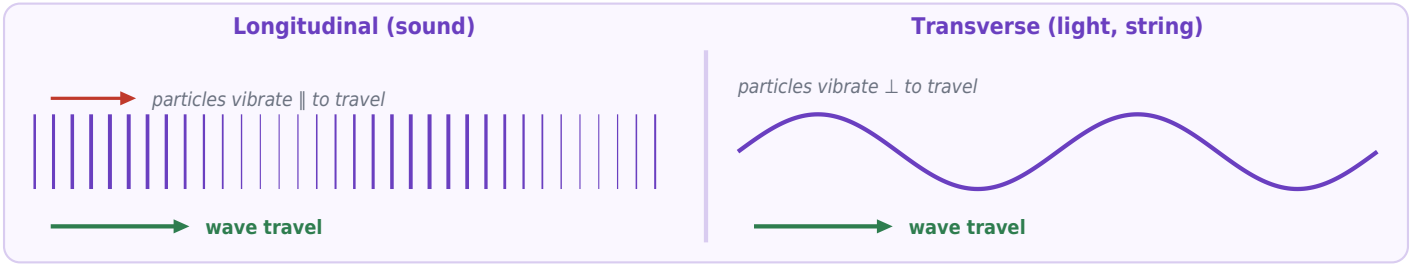
NCERT 10.3

COMPRESSION

Region where the particles are **pushed close together** (high density and pressure).

RAREFACTION

Region where the particles are **spread apart** (low density and pressure).

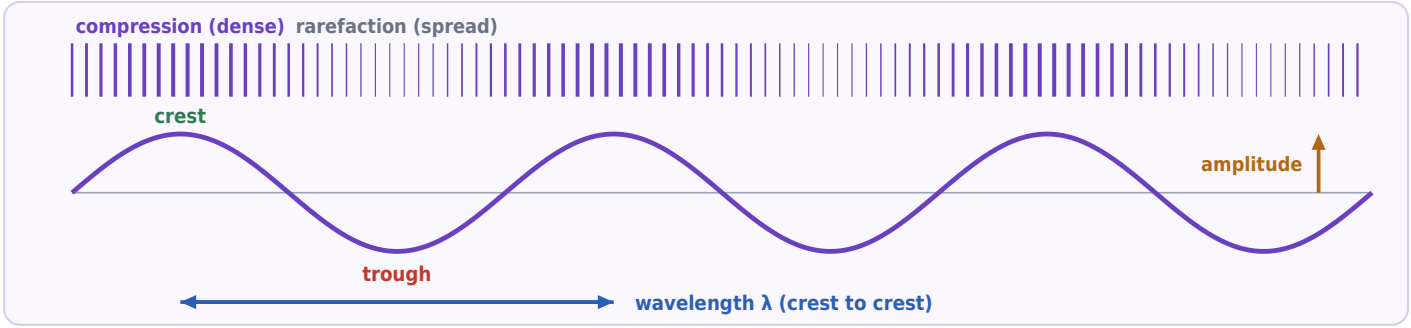


	Longitudinal wave	Transverse wave
Particles vibrate	parallel to the direction of travel	perpendicular to the direction of travel
Made of	compressions and rarefactions	crests and troughs
Examples	sound, slinky pushed along its length	light, ripples, a shaken rope

Mechanical vs non-mechanical: sound (and water waves) need a medium, so they are **mechanical waves**. Light does not need a medium, so it is a non-mechanical wave. A sound wave carries **energy** (a loud sound can shake windows) but not matter.

3 Describing a sound wave

NCERT 10.4

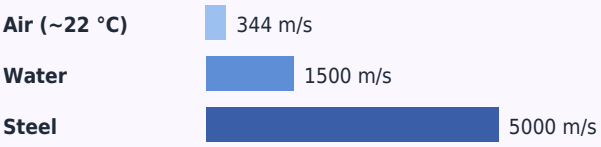


Term	Meaning	Unit
Wavelength λ	distance between two successive compressions (or crests)	metre (m)
Frequency ν	number of vibrations (waves) per second	hertz (Hz)
Time period T	time for one complete vibration · T = 1 ÷ ν	second (s)
Amplitude	maximum displacement from the rest position (height of the crest)	metre (m)
Speed v	distance the wave covers per second · **v = λ × ν**	m s^{-1}

v = λ × ν
speed = wavelength × frequency

★ MEMORY HOOK
Frequency fixed by the source. When a wave enters a new medium, its speed and wavelength change but its frequency stays the same.

Speed of sound in different media



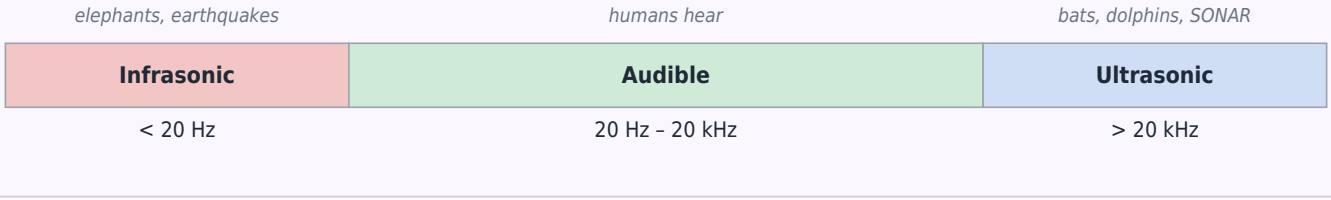
SPEED OF SOUND
Depends on the **medium** and its **temperature**. Air: 331 m s^{-1} at 0 °C, about 344 m s^{-1} at 22 °C; **higher temperature → faster**. Light (3 × 10^8 m s^{-1}) is far faster, so lightning is seen before thunder is heard.

4 How we perceive sound

NCERT 10.5

Quality	Depends on	High / low
Pitch (shrillness)	**frequency**	high frequency = high pitch (a whistle); low = bass (a drum)
Loudness	**amplitude** (and intensity)	bigger amplitude = louder; measured in decibels (dB)
Timbre (quality)	**mix of frequencies**	lets us tell a guitar from a violin playing the same note

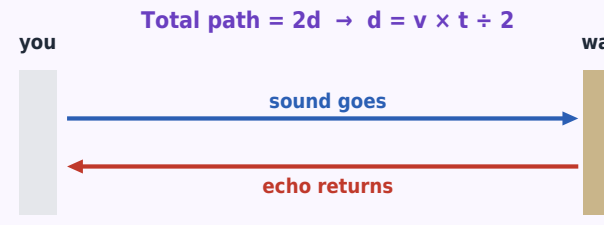
- **Intensity** = sound energy passing per second through unit area. Loudness is how our ear and brain judge that intensity. 0 dB is the faintest sound; very loud sound (above about 85–90 dB) can damage hearing.
- **Hearing:** sound enters the outer ear → eardrum vibrates → three small bones pass it on → the cochlea turns it into nerve signals for the brain.



5 Reflection of sound: echo

NCERT 10.6

Sound reflects from hard, rigid surfaces (wall, cliff, hill) and obeys the same laws of reflection as light. A reflected sound heard separately is an **echo**.



CONDITIONS FOR AN ECHO
- Our ear keeps a sound for **0.1 s**, so the echo must come back after **at least 0.1 s**.
- Minimum distance to the obstacle: d = 344 × 0.1 ÷ 2 ≈ **17 m**.
- If the echo returns in under 0.05 s it blends with the original: **reverberation** (a hall echoing, long rumbles).

Uses of reflection: megaphones and horns, stethoscopes, curved sound-boards in concert halls; halls use soft furnishings or curtains to reduce reverberation.

6 Ultrasound, infrasound and applications

NCERT 10.7

ULTRASOUND (above 20 kHz)

- **SONAR** measures depth of the sea, finds shipwrecks and submarines ($d = v \times t \div 2$).
- Medical **ultrasonography**, scans of organs and unborn babies.
- Cleaning delicate parts, finding flaws in metals.
- Animals: bats and dolphins use it for **echolocation**.

INFRASOUND (below 20 Hz)

- Made by earthquakes, volcanoes, thunder; used by elephants and whales to communicate over long distances.
- Some animals sense it before an earthquake.
- Humans cannot hear it, but it can be felt.

✎ Sonar (688.5 m)

Ultrasound returns from the sea-bed after 0.9 s in water ($v = 1530$ m s^{-1}): $d = 1530 \times 0.9 \div 2 =$ **688.5 m**.

✎ Solved problems

Revise, Reflect, Refine

Problem	Working	Answer
A source vibrates 5 times in 1 s. Find T and ν.	$\nu = 5$ Hz ; $T = 1 \div \nu = 1 \div 5$	$\nu = 5$ Hz, $T = 0.2$ s
$\nu = 20$ kHz, $v = 344$ m s^{-1}. λ?	$\lambda = v \div \nu = 344 \div 20000$	$\lambda = 0.0172$ m = 1.72 cm
Wave of 100 Hz, $\lambda = 17$ m. v? Distance in 10 s?	$v = \lambda\nu = 100 \times 17$; $d = vt$	$v = 1700$ m s^{-1} ; $d = 17000$ m
$T = 0.01$ s. Frequency?	$\nu = 1 \div T = 1 \div 0.01$	$\nu = 100$ Hz
Echo after 0.5 s ($v = 340$ m s^{-1}). Distance to the cliff?	$d = v \times t \div 2 = 340 \times 0.5 \div 2$	$d = 85$ m
Sonar signal returns after 4 s, $v = 1500$ m s^{-1}. Depth?	$d = 1500 \times 4 \div 2$	$d = 3000$ m

★ Quick revision

one-page summary

Quantity	Formula	Unit
Speed of a wave	$v = \lambda \times \nu$	m s^{-1}
Time period	$T = 1 \div \nu$	second (s)
Frequency	$\nu = 1 \div T$	hertz (Hz)
Echo distance	$d = v \times t \div 2$	metre (m)
Audible range	20 Hz to 20 kHz	
Speed in air	≈ 344 m s^{-1} at 22 °C (331 at 0 °C)	steel 5000 · water 1500

10 POINTS TO REMEMBER

1. Sound is produced by **vibrations** and needs a **medium**; it cannot travel in a vacuum.
2. Sound is a **longitudinal, mechanical** wave: compressions (dense) and rarefactions (thin).
3. Particles only vibrate about their position; the **disturbance and energy** travel.
4. λ = distance between successive compressions; ν = vibrations per second (Hz); $T = 1 \div \nu$.
5. **$v = \lambda \times \nu$**. The frequency is set by the source and does not change between media.
6. Speed: **solids > liquids > gases**; it rises with temperature.
7. **Pitch ↔ frequency · Loudness ↔ amplitude · Timbre ↔ mix of frequencies.**
8. Audible: 20 Hz–20 kHz. Below = infrasonic; above = ultrasonic.
9. An echo needs the reflector at least about **17 m** away (return time ≥ 0.1 s); $d = v\,t \div 2$.
10. Ultrasound: sonar, medical imaging, cleaning, echolocation in bats and dolphins.

✗ COMMON MISTAKE

- Using the full echo time in $d = v \times t$. The sound travels there **and back**, so divide by 2.
- Mixing up pitch and loudness: pitch = frequency, loudness = amplitude.
- Thinking air particles travel from the speaker to your ear. Only the disturbance does.
- Writing kHz values without converting (20 kHz = 20000 Hz).

TEST YOURSELF (cover the green column)

Q1. A wave has ν = 50 Hz and λ = 6.8 m. Find its speed.	**$v = 50 \times 6.8 = 340$ m s^{-1}**
Q2. An echo is heard 2 s after a shout (v = 340 m s^{-1}). How far is the cliff?	**$340 \times 2 \div 2 = 340$ m**
Q3. Which wave property decides pitch? Which decides loudness?	**Pitch: frequency; loudness: amplitude**
Q4. Why can't sound travel on the Moon?	**There is no medium (vacuum)**
Q5. Is a sound of 15 kHz audible? And one of 25 kHz?	**15 kHz: yes; 25 kHz: no (ultrasonic)**